Chapter 14

Managing for Environmental Justice

14.1 Introduction

Change. Complexity. Uncertainty. Conflict. Each requires attention in resource and environmental management. The people in Bangladesh, described in Chapter 1 and who are contending with the impacts of arsenic-contaminated water, are certainly encountering each of them. The individuals and groups in Pakistan who are striving to implement the National Conservation Strategy, which will balance economic, environmental and cultural considerations, are addressing them. The *precautionary principle* was created as a guideline to handle them. *Hedging* and *flexing* have been recognized as alternative approaches toward them. *Visioning*, *backcasting* and *adaptive environmental management* are being increasingly used to anticipate and to respond to them. Techniques such as *Life Cycle Analysis* and *Environmental Audits* have been created to help managers address

Box 14.1 A Jeremiah* parable

I imagine spaceship earth as a kind of fortunate Titanic. On the ship's prow in the middle of the night, Jeremiah Jones peers into the dimness. Faintly perceiving some ominous shapes ahead, he cries out lustily "icebergs ahead." Unsure if he is heard, he cries out again and again. On the ship's bridge, the captain, hearing Jeremiah only after some time, turns to the navigator and asks for a course correction to avoid a collision. Ten degrees to the starboard she says. The Captain, thinking "What luck that I have already started to turn because of the bad weather ahead'" orders a five degree correction. The helmsman looks at his compass and suddenly realizes that he has been dozing for a few minutes and that the ship has actually been drifting, fortunately in the right direction. Without saying anything, he then corrects the course by two degrees. Up ahead, alone and in the cold, Jeremiah awaits a hard starboard course correction, maybe even a reversal of engines. Sensing none, he mutters to himself "they never listen to me" and prepares for the worst.

(* The prophet Jeremiah is the archetype of those who warn of future woes, hence the term *jeremiad* as a tale of woe.)

Source: Kates, 1995: 635 and 625.

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them. And, increasing attention to *implementation*, and to *monitoring* and *evaluation*, explicitly acknowledge the need to be able to modify approaches as a result of them.

As mentioned throughout this book, it is often presumptuous to believe that humans "manage" environment and resources. More realistically, humans manage their interactions with environment and resources. For that reason, uncertainty (as a result of imperfect understanding), as well as ignorance, and conflict (as a result of many different and legitimate interests) are common. Furthermore, the complexity of biophysical and socio-economic systems is great, and is exacerbated by ongoing change, some of which is influenced or caused by human activity. Thus, to develop policies, programmes or plans for resource and environmental management, or to be able to appraise the effectiveness of initiatives, it is important to be able to recognize and deal with change, complexity, uncertainty and conflict.

In this chapter, discussion focuses on four matters deserving systematic attention: (1) developing a vision, (2) creating a process, (3) generating a product and (4) ensuring implementation and monitoring.

14.2 Developing a vision

In Chapter 3, reference was made to a comment by Alice in Wonderland – if you do not know where you want to go, any road will get you there. In other words, if there is no sense of vision or direction regarding a desirable future, then almost any choice will do. In contrast, if we have a sense of where we would like to get to, then it should be possible to take actions or intervene to try and move in the desired direction.

Given the above comments, a key task for resource and environmental managers is to help develop a vision for a desired future, as considered in Chapters 3 and 4. This task is never easy, since societies rarely if ever are homogenous (Box 14.2). Many interests exist, and conflicts frequently emerge. Thus, what may be a desirable future for one group or interest may be viewed as undesirable by another. However, by considering *desirable* futures, we should be able to move away from what too often is a preoccupation with *most probable* futures. If future search conferencing or backcasting is to be used, we need a vision of

Box 14.2 How far to look into the future?

It is with great trepidation that one writes about the future. After all, it hasn't happened yet and one could end up looking pretty silly, depending upon the acuity of one's vision. For that reason, and like many other forecasters, I have adopted the convention of choosing as the end of my forecast horizon a date sufficiently far in the future that there is no chance of my being around to be held accountable.

Source: Portney, 2000: 6.

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Box 14.3

It is extremely important to approach sustainability as a learning process rather than as series of blueprints.

Source: Brooke and Rowan, 1996: 115.

where we want to get to, so that decisions can be made which take us in the prescribed direction.

Since many different interests normally coexist in a society, determining a vision is not a task only for professional resource and environmental managers. Consultation and interaction with people living in an area are important, in order to draw upon their knowledge and understanding of conditions and processes in that area, to address their needs and to incorporate their expectations. For this reason, *partnerships* and *participatory approaches* should be important features of resource and environmental management, with particular need to draw upon *local knowledge systems*. Furthermore, in developing partnerships, greater attention needs to be given to achieving *gender balance*.

Since publication of *Our Common Future* in 1987, *sustainable development*, today more usually referred to as *sustainability*, has been viewed by many as the vision to be pursued. As explained in Chapter 4, the World Commission on Environment and Development did not present one blueprint for sustainable development, arguing instead that different countries and cultures have to custom-design an individual strategy reflecting their conditions and needs (Box 14.3).

The outcome is that different interpretations have been given to sustainability in various countries, particularly between developed and developing nations. Such differences or inconsistencies are not inherently bad. Indeed, Ralph Waldo Emerson wrote that foolish consistency is the hobgoblin of little minds. Various interpretations of sustainability are appropriate if they reflect the different situations in countries or regions. Furthermore, within a country or region, the emphasis in a sustainable development strategy could reasonably be expected to evolve over the years as conditions change. Ongoing interest in sustainable development, as a vision, can be expected to continue because it provides a way to consider how to balance economic, environmental and cultural matters. The guest statement from Walter Danjoy provides a perspective about sustainability related to developing countries.

Another concept increasingly drawing attention as a possible vision is *environmental justice*. Higgins (1993: 292) has observed that, regarding environmental justice, ". . . advocates argue their communities are disproportionately burdened with polluting facilities, the beneficiaries of which live largely outside of the minority community sites". More generally, Cutter, Holm and Clark (1996: 517) stated that "[environmental] inequity originates from three major sources of dissimilarities: social, generational and procedural. To test for outcome equity, one examines the disproportionate effect of environmental degradation on places or people arising from these dissimilarities".

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Guest Statement

Sustainable development in developing countries

Walter Danjoy, Peru

Introduction

Within the context of sustainable development, developing countries, such as Peru, find sharply contradictory situations, which require a more realistic evaluation with respect to the existing relationship between the availability of natural resources, their use by the population in general and particularly by native and rural communities, and the concept of sustained development. The classic concept of sustained resource management, which is aimed at resource preservation in order to meet the needs of future generations, finds a barrier in the social element. This barrier, which limits sustainable development, is determined by the urgent needs of the population of the areas in which the resources and their area of influence are located, giving rise to the need for specific strategies for the use of these resources.

Nature of the problem

In developing countries, economic problems are frequently both a symptom and a cause of environmental degradation. Economic necessity leads to overgrazing and deforestation, thus reducing the productivity of land and soil, increasing the frequency of floods and droughts, causing greater poverty and despair, particularly in rural underdeveloped sectors of a country. The poor become trapped in a downward cycle of rapid population growth, environmental degradation and falling economic standards. They are unable to grow economically because of the pressure of population, lack of agricultural land and the inadequate use of natural resources as a source to contribute to the active national economy. Many developing countries find it difficult to overcome these controlling economic factors.

The synergistic interaction of poverty, illiteracy, malnutrition and underdevelopment make certain problems severe. Examples of this are high infant mortality, cholera and mycotoxins in food. Any improvement in this situation needs concerted action in more than one activity, particularly in the sustainable use of natural resources by the population.

Conflicts of use

The sustainable development and persistent productivity concepts may become compatible with those related to the growing need to increase food supply, minimizing the conflict among production, sustainable systems and environmental preservation. To be able to apply practices which will guarantee a sustainable production system, it is essential that they must be profitable in order to fund the investments required for the preservation of the productive environment. If adequate technological elements and strategies are not applied, combined with an element of profitability, it may turn out that the increased production may negatively affect the natural resource on which it is based, creating a vicious circle which would be hard to break.

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When an economic depression hits, the rural population is forced to intensify the use of its resources. In many cases, rural inhabitants are blamed for the degradation of natural resources, mainly because of their cut-and-burn practice and migrating-type of agriculture, which is recognized as the paradigm leading to the destruction of the Amazon environment. However, there is some evidence that, in most cases and under economic and social stability conditions, rural residents may turn into a dynamic agent participating in the process, carrying out a series of strategies for the protection of their resources. When the country's economic conditions degrade, producers and rural residents change their production practices and strategies, perhaps speeding up the process of environmental degradation.

Finally, within the context of the management of natural resources and sustainable development, the above-mentioned situation creates the need for governments to adopt integrated social policies aimed at aiding the marginal population, who suffer a reduction of their only natural and cultural heritage, particularly when an economic recession alters economic levels; and consequently, the potential of natural resources.

Walter A Danjoy was born in Huancayo, Peru. Beginning at the age of 12, he came in contact with the Amazon people while accompanying his father on trips to the Upper Jungle. In 1972, he graduated from the National University of Central Peru as a forest engineer; and after graduation began his professional career at the National Office for Natural Resources Evaluation, where he worked for 20 years, carrying out activities related to the inventory and evaluation of natural resources and environmental management and monitoring, where he applied geographic information and remote sensing systems. In 1992, he completed a Master's degree in Canada, with a thesis focused on the role of remote sensing and geographic information systems for management of natural resources in the Peruvian Amazon Basin.

Currently, he is a consultant for Peruvian and international agencies, on issues related to the management of natural resources and environmental monitoring. He is also a University Professor and author of four books on natural resources. He is a member of the Latin American Society of Specialists on Remote Sensing and Geographic Information Systems (SELPER), of which he was President between 1992 and 1994.

The environmental justice movement combines values and goals from the civil rights, poor peoples, occupational safety and health, and grassroots environmental movements, and "the vision that emerges is a qualitative challenge to existing business and government agendas, as well as to the reform agendas of mainstream environmentalism" (Higgins, 1993: 292). These challenges emerged in the late 1970s and early 1980s, and focused on institutional responses related to toxic waste disposal, public facility siting (from airports to electric power transmission lines), species and wilderness preservation, nuclear power and weapons, and the political positions and growing professional orientation of mainstream environmentalism.

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Those working within the framework of environmental justice do not seek only to redistribute existing and future pollution. More fundamentally, they advocate policies and practices which incorporate ecologically, economically and socially appropriate development, and argue that pollution *prevention* rather than pollution *control* should be a primary goal. The emphasis on prevention rather than control has the environmental justice movement directly opposed to strategies based on control of pollution, usually reflected in market-based approaches (permits to pollute, user pay).

Another defining characteristic of environmental justice is that it advocates *empowerment* as an end in itself, as well as a means to achieve substantive improvements. For environmental justice activists, government agencies are viewed as being too frequently uncommunicative, unresponsive, disdainful and openly hostile (Higgins, 1993: 292). As a result, environmental justice advocates promote open, transparent and accessible decision-making processes. In that context, they also believe an important task is to ensure that technical and/or scientific documents should be "deciphered and translated" so as to be understandable for citizens who wish to become informed about environmental issues, and to participate in environmental planning and decision making.

Environmental justice is a very broad concept, and incorporates numerous dimensions. Box 14.4 provides definitions for environmental justice, and several

Box 14.4 Definitions of key terms

Environmental justice: The right to a safe, healthy, productive and sustainable environment for all, in which "environment" is viewed in its totality, and includes ecological (biological), physical (natural and built), social, political, aesthetic, and economic components. Environmental justice refers to the conditions in which such a right can be freely exercised, through which individual and group identities, needs, and dignities are preserved, fulfilled, and respected in a way which provides for self actualization and personal and community empowerment.

Environmental equity: An ideal of equal treatment and protection for various racial, ethnic and income groups under environmental statutes, regulations and practices applied in a manner that yields no substantial differential impacts relative to the dominant group, and the conditions so created. It implies concepts of "fairness" and "rights", but does not necessarily address past inequities.

Environmental racism: Involves racial discrimination in environmental policy making, enforcement of laws and regulations, and targeting of communities of colour for noxious waste disposal and siting of polluting industries. Racial discrimination can be intentional or unintentional, and often reflects "institutional racism".

Environmental classism: The results of and the process by which implementation of environmental policy creates intended or unintended consequences which have disproportionate impacts (adverse or beneficial) on lower income persons, populations, or communities.

Source: Modified from: http://www-personal.umich.edu/~jrajzer/nre/definitions.html

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associated key concepts. From Box 14.4, it is apparent that we need to differentiate among environmental justice, equity, racism and classism.

A decision about a land-fill site in Warren County, North Carolina, during 1982 is generally recognized as the catalysing event for the environmental justice movement in the USA. A decision was taken to create a land-fill site to receive polychlorinated biphenyl (PCB) contaminated soil, removed from 14 different places in the state, adjacent to a small, low-income, primarily African-American community. This decision highlighted that such noxious facilities were often being located in areas inhabited by minorities and/or low-income people. Civil and states rights activists organized many demonstrations, resulting in the arrest of over 500 individuals. Subsequently, the US General Accounting Office conducted a study of eight states in the US South to examine the correlation between siting of hazardous waste disposal sites and the racial and economic status of surrounding communities. The results of the study indicated that three out of every four such land fills were located in or near minority communities. Other studies confirmed these findings. From the time of the Warren County decision, the concepts of environmental justice, equity, racism and classism started to appear in public policy discussions and the research literature. These concepts can be expected to be increasingly important elements of visions in the future.

14.3 Creating a process

In the 1990s, the Ontario Ministry of Natural Resources undertook a review of its planning process. As the person responsible for the review explained, the process had to be modified so that it was not the main "issue" when resource allocation decisions were made. In other words, the official meant that too often the Ministry of Natural Resources was finding that people were not satisfied with decisions because of concerns about weaknesses in the planning process. Ministry of Natural Resources staff understood that it was not possible for every resource allocation decision to satisfy every interest, given the many conflicts. However, they did believe that an improved process should be able to result in those who did not get what they wanted being satisfied that the planning process had been fair and reasonable.

The experience in Ontario emphasizes therefore that a vision for resource and environmental management requires an accompanying process to identify issues and problems, assemble necessary information and viewpoints, determine alternative solutions, and select a course of action. And, increasingly, citizens are expecting to be involved in such planning and management processes, highlighting the importance of *partnerships* and *local knowledge systems*. And, as mentioned in Section 14.2, which focused upon development of a vision, there is growing awareness regarding the need to ensure greater *gender sensitivity* in the construction of partnerships. One form of partnership that has attracted increasing attention is *co-management*, in which specific roles and powers are delegated to local people. Co-management has been particularly effective in incorporating local knowledge systems.

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The complexity and uncertainty associated with natural and human systems also have encouraged the development of what have become known as *learning organizations* and an *adaptive environmental management* process, or AEM. AEM explicitly recognizes that planners and managers deal with turbulent conditions, and that they can expect to be surprised on a regular basis. As a result, in this approach, planning and management are designed to encourage learning from mistakes, and to facilitate ongoing adjustments. AEM is realistic in its recognition of complexity and uncertainty, and of often rapidly changing conditions. However, many planners and managers still have difficulty acknowledging when they are mistaken or wrong. Furthermore, many clients or constituents are unlikely to be impressed by plans and management strategies which are continuously being changed, as that makes it difficult to know what investment decisions to make. Thus, while the concept of AEM is in many ways realistic, it also contains much idealism which often will be resisted in the "real world".

The presence of conflict has led to the emergence of *alternative dispute resolution* (ADR) processes. As indicated in Chapter 11, ADR is not a panacea for resolving all conflicts. Instead, it is an alternative to the more established judicial or legal process. The judicial system can be unduly adversarial and expensive, and lead to winners and losers. In contrast, ADR seeks to create a process in which parties in conflict come together voluntarily for joint problem solving. When parties are prepared to address their conflicts on a voluntary and joint basis, ADR offers an attractive alternative. However, if the various sides in a dispute are too entrenched in their positions and not interested in looking for mutually beneficial solutions, then ADR is not likely to be helpful.

Another process consideration relates to the role of *interdisciplinarity* in research and problem solving (Box 14.5). In this book, one message is that the perspectives and insights of many disciplines and professions are required. However, while an interdisciplinary approach is often advocated for resource and environmental management, the concept is interpreted in many different ways. When a research team or management group contains individuals using interdisciplinarity in different ways, it is unlikely that the full potential of the approach will be realized.

As Klein (1990: 56–7) noted, most activities presented as "interdisciplinary" are not, but instead are "multi-disciplinary". The distinction between the two is based on the way in which integration and synthesis are pursued. A multidisciplinary approach is designed to take advantage of the expertise of specialists in various disciplines. With regard to sustainable development, for example, a

Box 14.5 Interdisciplinarity

Two claims about knowledge appear widely today. The first claim is that knowledge is increasingly interdisciplinary. . . . The second and related claim is that boundary crossing has become a defining characteristic of the age.

Source: Klein, 1996: 1.

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multi-disciplinary approach might involve an ecologist, economist, sociologist and other disciplinary experts each posing the questions judged to be important in their discipline, and then conducting separate analyses, drawing upon the frameworks, perspectives, concepts and methods of their respective disciplines. They would work on their own and complete separate reports. Once the various reports were completed, one or more people would use the information and insights from the reports to try and obtain understanding which would be broader and deeper than from any one discipline. Thus, integration and synthesis occur at the end of the process of analysis.

In contrast, in an interdisciplinary approach, the integration and synthesis begins at the outset of the process, and continues throughout. In a sustainable development project, the various disciplinary specialists would meet before any data collection or analysis had started. The first task would be to share views about the nature of the problem to be examined, to clarify which contribution each discipline could make, and to indicate what assistance could be provided from one discipline to others. Once that task was completed, the disciplinary specialists would conduct their individual studies, but throughout the research would regularly meet with counterparts in other disciplines to share data and insights, and to discuss issues and problems for which assistance was needed. An interdisciplinary approach is more time consuming, as it requires commitment from disciplinary specialists to meet with each other on a regular basis. Furthermore, an interdisciplinary approach can be threatening to some, since specialists from other disciplines are more likely to question fundamental beliefs and assumptions that any one discipline accepts as part of its culture. However, if effective communication and sharing can occur among the team members, then one important end product of an interdisciplinary approach, beyond the systematic and continuous synthesis, is creation of respect, good will and trust among people from different disciplines. When this happens, and individuals are able and willing to acknowledge the limitations and weaknesses of their discipline, the conditions have been established for more collaborative interdisciplinary work in the future, work likely to be of even higher quality as a result of a willingness of specialists to listen to and incorporate views of specialists from other fields.

Interdisciplinarity is not a perfect approach, and indeed the obstacles to build up the necessary respect, goodwill and trust are formidable. Obstacles include the sensitivities and egos of participants, and perceived relative status or prestige of different disciplines. Sometimes an interdisciplinary team cannot overcome such barriers. When this occurs, the team may refer to itself as conducting interdisciplinary research, but in fact is engaged in multi-disciplinary work.

Another legitimate concern may be that as specialists from one discipline borrow concepts from another discipline, the result may be misuse or abuse of the concepts. In this regard, Klein (1990: 88) observed that six common problems can occur when concepts from one discipline are borrowed by another: (1) distortion and misunderstanding of the borrowed material; (2) theories, concepts, methods and data used out of context; (3) the borrowed material used in one discipline is out of favour in its home discipline; (4) phenomena treated cautiously or with scepticism in their home discipline accepted as givens or as

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"certain" by the borrowing discipline; (5) a particular theory or perspective is over-used or over-relied upon; and (6) the borrowing discipline over-looks or dismisses contradictory evidence, tests and explanations. These limitations are provided here, to help avoid an uncritically enthusiastic pursuit of interdisciplinarity.

For resource and environmental management, whatever process is used does not end with the development of a strategy or a plan. As noted in Chapters 12 and 13, it is also important to created processes to facilitate implementation and monitoring. These are considered in more detail in a later section.

14.4 Generating a product

Careful thought in the crafting of a vision and a process is essential. However, planners and managers should never forget that the ultimate purpose is to resolve a problem or create an opportunity for positive change. As a result, a vision and process should lead to an output, which could be a strategy or plan, and then ultimately to outcomes and impacts, or new desirable conditions as a result of initiatives flowing from a strategy or plan. As we became aware when examining *learning* organizations and *adaptive environmental management*, a strategy or plan should not be viewed as a static product. In other words, the preparation of a strategy or plan does not mean that the work of resource and environmental planners and managers is completed. Due to changing conditions, turbulence and surprises, it may become necessary to modify the product. Furthermore, as a result of new understanding and knowledge, planners and managers may become aware of opportunities for improvement. A key lesson is that the product should never be viewed as fixed and unchangeable.

The design of the strategy should incorporate some basic ideas. First, it should reflect the spirit of the *precautionary principle*. In other words, lack of knowledge or understanding should not be used as a reason for not taking action. Second, in situations characterized by high uncertainty, it is appropriate to base the approach on *flexing*. In other words, the approach should strive to achieve what is considered to be the option providing the most benefits, but also recognize that there may be a need to make adjustments as time unfolds.

In developing a product which can guide actions to resolve a problem or to create opportunities, planners and managers should draw upon a mix of techniques or tools. In this book, it has been argued that in the early stages we should use a combination of *forecasting* and *backcasting* to ensure we consider the most probable, desirable and feasible futures. Having outlined some options, we then need to analyse them systematically. Such analysis should use various techniques. *Environmental impact assessment* should draw attention to the environmental gains and losses of possible choices. And, if impact assessment is broadened to incorporate *social impact assessment*, then sociocultural issues also will be systematically considered. *Life-cycle analysis* emphasizes that we should consider the implications of options from "cradle to grave". Finally, *environmental audits* provide another way to determine performance relative to some generally accepted standards or guidelines.

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Resource and environmental planners and managers have a broad suite of techniques from which to choose when analysing options along the way to developing a strategy or plan. As in all good analysis, it is important not to over-rely on any one method or technique, but to use a combination to ensure that the strengths of one offset the weaknesses of another. Judgement has to be exercised about the number of techniques to be used, since the addition of each extra technique implies more time and resources to complete the task. At some point, diminishing returns set in, and the advantages of conducting one more type of analysis becomes marginal. There is no recipe book which can identify the most appropriate mix of methods to use. Analysts have to make choices, based on their judgement about conditions and needs.

A continuing theme in this book is that the resource and environmental management process never ends. Nothing highlights that more than the concepts of implementation and monitoring. Visions, processes and products are unlikely to have an impact if action does not occur, and there is the capacity to make adjustments. It is these two aspects, implementation and monitoring, which are considered in the next section.

14.5 Ensuring implementation and monitoring

In Chapter 12, it was noted that many obstacles can thwart effective implementation of policies, programmes or plans. Indeed, it was commented that implementation failure seems to be similar to "original sin". Both appear to be everywhere and to be ineradicable! And, evidence would seem to support this conclusion, since there often seem to be more plans generated than action taken.

The discussion in Chapter 12 identified many obstacles which can hinder implementation. We also became aware that there are *programmed* and *adaptive* approaches to implementation. The conclusion in that chapter was not that one approach was correct and the other was wrong, but rather that each offers advantages in different situations. As planners and managers, we should be aware of the strengths and weaknesses of programmed and adaptive approaches, and be able to determine which one is most likely to be effective for a given situation. In Chapter 2, we also reviewed the merits of different *planning models* or *schools*. These make different assumptions about problem situations. Regarding implementation issues, it would seem that more use of *incremental* and *transactive* planning models might lead to improvements.

A wise person learns from his or her, or other people's, mistakes. To be able to learn from experience requires a willingness and capacity to track or monitor. Certainly, monitoring is one of the cornerstones of *learning organizations* and *adaptive environmental management* which approach problem solving as a social learning experiment, and deliberately seek information which will allow improvements to be made. In Chapter 13 we saw that many agencies and countries have introduced *state of environment* or *state of sustainability reporting* as a way to monitor policies, programmes and plans. *Environmental auditing* is also used, often at a site level, to measure performance relative to some pre-determined

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Figure 14.1 Children in the village of Tumba, Sokoto State, in northwestern Nigeria. Decisions taken by resource and environmental planners and managers will affect their lives, as well as generations to follow (Bruce Mitchell)

standards. Much monitoring is still at a fairly basic level, with attention being given more to *inputs* than *outputs*, mainly for the pragmatic reason that inputs usually are easier to measure and collect data on. Hardly any attention is given to monitoring outcomes and impacts, which occur in the medium to long term.

It is difficult to conceive how we can learn from experience or mistakes if we do not systematically monitor our initiatives and activities. Yet, the reality is that often there is considerable resistance to monitoring. People who have invested professional reputations or egos in policies or plans often are not anxious to see them monitored, if there is significant likelihood that errors will

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then become public knowledge. This reluctance and inertia in many instances still has to be overcome, and this will not be a minor task.

14.6 Anticipating the future

Will we encounter the future like a Captain Ahab travelling on the *Pequod* in pursuit of Moby Dick, or like a Jeremiah Jones travelling on a "fortunate" *Titanic* (Box 14.1)? A main message from this book is that we do have choices; it is up to us to identify what those are and how we might act on them. The future may appear unclear and fuzzy, and conflicting signals may be seen regarding the most appropriate course of action to pursue. Furthermore, we must be mindful that the decisions we take today will have implications for our children, and future generations (Figure 14.1).

While the "right" course of action may not be obvious, decisions still must be taken, and commitments made since needs exist, problems emerge or persist, and opportunities present themselves. In that context, we should heed the words of Emil Salim (1988: v), the first Minister of Population and Environment for Indonesia and a member of the World Commission on Environment and Development. While he was speaking about his own country, his comments have much more general relevance:

> As a developing country, Indonesia faces the necessity of having to start sailing while still building the ship. We don't have the time to wait until all concepts are well established; until the theories are completed. The problems cannot wait until we can think the problems through.

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